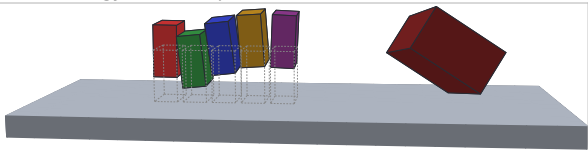
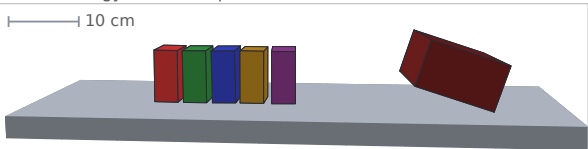


(a) mass-shaped row weight $w = 1/M_q$ dashed: the books at rest
board-mode energy 1,707 J peak book rise 56.8 mm



(b) matched row weight $w = 1/(4M_q + h^2K_q)$
board-mode energy 0.35 J peak book rise 5.3 mm



1 iteration x 8 substeps per 1/120 s frame, modal relaxation 1.0, true scale
passivity governor off, two independent runs, 114 ms after impact
steel cantilever shelf, 6 kg dropped 1.00 m, never within 125 mm of a book